

Model Evaluation

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 - Accuracy
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Confusion matrix - bu classification modelning natijalarini boshlab uchun ishlatiladigan jadval. U modelning nechta to'g'ri va nechta noto'g'ri bashorat qilganini ko'rsatadi.

y-actual	y-pred	sample
1	1	1
0	0	2
1	0	3
0	0	4
1	1	5

0 - Negative

1 - Positive

Caption

	Predicted 0	Predicted 1
Actual 0	True Negative	False Positive
Actual 1	False Negative	True Positive

	Predicted 0	Predicted 1
Actual 0	2	0
Actual 1	1	2

from sklearn.metrics import confusion_matrix,
classification_report

y_actual = [1, 0, 1, 0, 1]
y_pred = [1, 0, 0, 0, 1]
cm = confusion_matrix(y_actual, y_pred) ✓

array([[2, 0],
[1, 2]])

Accuracy - modelning nechta baho-
ratni to'g'ri qilganini ko'rsatishi-
ni oddiy baxolash metrikasi.

$$Acc = \frac{True\ Positive + True\ Negative}{True\ Pos + True\ Neg + False\ Pos + False\ Neg}$$

$$Acc = \frac{2 + 2}{2 + 2 + 0 + 1} = \frac{4}{5} = 0.8 \cdot 100\% = 80\% \checkmark$$

Caption

Precision - model Positive deb prediqt qilgan natijalar ning qanchasi haqiqatdan ham Positive ekanligini ko'rsatadi.

$$\text{Precision} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Positive}}$$

$$\text{Precision} = \frac{2}{2+0} = 1 \cdot 100\% = 100\%$$

Recall - model haqiqiy Positive holatlar ning qanchasini to'g'ri topdi qanini ko'rsatadi.

$$\text{Recall} = \frac{\text{True Positive}}{\text{True Positive} + \text{False Neg.}}$$

$$\text{Recall} = \frac{2}{2+1} = \frac{2}{3} \approx 0.67 \rightarrow 67\%$$

Precision va Recall farqi

Precision: Model Positive deganlar ning qanchasi to'g'ri?

Recall: Haqiqiy Positive lar ning qanchasini model topdi?

Recall - modelning haqiqiy Positive holatlar ni topish qobiliyatini o'lchaydi.

Caption

F1-Score. - bu Precision va Recall ning muvozanatli (harmonik o'rtacha) ko'rsatkichi.

$$F1 = 2 \cdot \frac{\text{Precision} \cdot \text{Recall}}{\text{Precision} + \text{Recall}}$$

$$F1 = 2 \cdot \frac{1 \cdot 0,67}{1 + 0,67} = 0,88 \rightarrow 88\%$$

Classification Report. - bu classification modelning ishlashini batafsil hisobot bo'lib, unda Precision, Recall, F1-Score va Support ko'rsatkichlari har bir sinf uchun alohida beriladi:

Logistic Regression:
Logistic Regression ---- Sigmoid function (logi

$$\sigma = \frac{1}{1 + e^{-z}} \quad e = 2.71$$

$$z = d + d_0 \cdot x_0 + d_1 \cdot x_1 + \dots + d_n \cdot x_n$$